KARNATAKA STATE COUNCIL FOR SCIENCE AND TECHNOLOGY

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49th SERIES OF STUDENT PROJECT PROGRAMME

List of Broad Themes / S&T technologies for submitting the project proposals

The **Karnataka State Council for Science and Technology (KSCST)** is initiating the **49th Series of the Student Project Programme**, commencing in **October 2025** for the **2025-26 academic year**. This program offers a significant opportunity for final-year engineering students to showcase their innovative projects.

Eligibility and Submission Details

KSCST invites project proposals from final-year students in the following programs for sponsorship under the **49th Series of the Student Project Programme (SPP)**:

- **BE**
- **B. Arch.**
- **M.Tech.**
- **M.Sc.**
- **M.Sc. (Agriculture)**
- **MBA**
- **MCA**

Submission Period: Proposals can be submitted through Google Form from **3rd October 2025** to **15th December 2025**.

Thrust Areas for Project Proposals

The submitted projects must align with one of the following broad indicative themes:

1. **Technologies for Rural Livelihood Improvement:** Focus areas include energy, water, agriculture, environment, health, livelihood, green technology, assistive aid, handicraft, textile, animal health, post-harvest, and education.

SL NO.	TECHNOLOGY/ PRODUCT DETAILS	FOCUS AREA
i.	• (EDU-BOX) SOLAR PROJECTOR AND OFFLINE SERVER. • SOLAR POWERED SHEEP SHEARING MACHINE • SOLAR WATER FILTRATION AND VENDING MACHINE	ENERGY, WATER
ii.	• LOW-COST SENSOR AND DEEP LEARNING-BASED METHOD TO SPRAY THE FARMS WITH CONTROLLED OR LIMITED PESTICIDES WITH EQUAL DISTRIBUTION	AGRICULTURE, ENVIRONMENT
iii.	GREEN POLYMER NANOCOMPOSITE (BIOPLASTIC) FROM AGRICULTURAL RESIDUE FOR FOOD PACKING APPLICATION	

iv.	• IMPROVED HAPPY SEEDER TECHNOLOGY FOR PADDY STRAW MANAGEMENT ON FARMER'S FIELD	
v.	INTEGRATED BIOTECHNOLOGICAL APPROACH FOR THE TREATMENT OF MICROPLASTICS (MPS) AND PLASTIC DERIVED COMPOUNDS (PDCS) ALONG WITH REFRACTORY ORGANICS IN THE MUNICIPAL LANDFILL LEACHATE	ENVIRONMENT
vi.	ENERGY EFFICIENT MULTIFUNCTIONAL FIELD DISTILLATION UNIT FOR ISOLATION OF ESSENTIAL OILS	ENERGY
vii.	• MOUNTAIN SOLAR WATER AND SPACE HEATING SYSTEMS • HERBAL CHIRAYITA FORMULATIONS FOR CONTROL OF DIABETES, ANTI-BACTERIAL, ANTIVIRAL, ANTI- FIBROID AND HEPATO-PROTECTIVE ACTIVITIES. • SOLAR BIOGAS HYBRID SYSTEMS FOR MOUNTAIN NEEDS.	ENERGY, WATER, HEALTH
viii.	• RESILIENT FARMING TOOLS & TECHNOLOGIES IN FLOOD AND WATER-LOGGED AREAS (CYCLE WEEDER AND HOEING, SOIL DIGGER, PADDY TRANSPLANTING, IMPROVISED PORTABLE LOW TUNNEL POLY HOUSE) AND POST-HARVEST TOOLS (TURMERIC RAMMER TOOL, GROUNDNUT DE-HUSKING TOOL, TURMERIC STEAM BOILER, ROOT CROPS WASHING TOOL, ZERO ENERGY COOL CHAMBER, MAIZE DE-HUSKING MACHINE, BRIQUETTE MAKING TOOL) FOR REDUCING LABOURER COST	AGRICULTURE, ENERGY
ix.	• COST-EFFECTIVE BRIDGE (BAMBOO, WOODEN & STEEL) FOR RURAL CONNECTIVITY • UPGRADED WATERMILLS WITH LOW HEAD TURBINE DESIGN AND STANDARDIZATION	LIVELIHOOD, WATER
x.	• IMPROVED BIOMASS COOKSTOVES • PV INTEGRATED SOLAR TUNNEL DRYER • SOLID STATE BIO-GAS PLANT • KITCHEN WASTE-BASED BIO-GAS PLANT • WALK IN TYPE SOLAR TUNNEL DRYER • DHABHA SIZE IMPROVED BIOMASS COOKSTOVES • FLUIDIZED BED GASIFIER • MSW ORGANIC WASTE BIO-METHANATION • SOLAR-BIOMASS BASED HYBRID DRYER	GREEN TECHNOLOGY, ENERGY,
xi.	• ARID SPECIFIC TREE BASED FARMING SYSTEM (WADI) • BREED IMPROVEMENT OF MARWARI SHEEP AND GOAT • DOORSTEP AI SERVICES FOR ARTIFICIAL INSEMINATION AND PREVENTATIVE HEALTH CARE OF CATTLE. • ESTABLISHMENT OF CACTUS AS A FODDER (NEW CROP) • LOW-COST GRAVITY-BASED DRIP IRRIGATION SYSTEM AND EFFICIENT WATER USE TECHNIQUE • TANKA WITH AGOR AND ROOFTOP RAINWATER HARVESTING FOR TAPPING OF RAINWATER	AGRICULTURE, LIVELIHOOD, INFORMATION TECHNOLOGY, WATER
xii.	• BIOMASS DRYERS, PYRO LITE STOVE, OUTDOOR ANIMAL FEED STOVE, • VERSION 2 OF PYRO STOVE WITH NEW INSULATING MATERIALS AND MANUFACTURING IMPROVEMENTS • 2 VERSIONS OF FORCED DRAFT AGNI SAKHI & AGNI MITHRA STOVE • FUEL EFFICIENT BAKERY OVEN • CUSTOMIZED SARALA STOVE DESIGNS	LIVELIHOOD, ENERGY
xiii.	• BIOMETRIC SOLAR ATM NAMED AS 'JANATA ATM' • MICRO SOLAR DOME • MICRO SOLAR PUMP COUPLED WITH RO SYSTEM &	ENERGY, GREEN

	INTEGRATED WITH PRE FILTER AND INTELLIGENT CONTROLLER • MECHANIZED FERRO CEMENT BASED PRODUCTS • BAMBOO BASED SHUTTERS AND DOORS FOR RETROFITTING OF HOUSES	TECHNOLOGY, LIVELIHOOD
xiv.	• NECK REST PILLOW WITH MEMORY USE • BRICK MAKING FROM GRANITE WASTE • ERGONOMIC DESIGN OF MECHANICAL WEEDING UNIT FOR REDUCING PHYSICAL STRAIN • E-WASTE SEGREGATION AND PREPROCESSING MODEL • CUSTOMISED GIFT MAKING THROUGH 3D CHOCOLATE PRINTING	LIVELIHOOD, ASSISTIVE AID, GREEN TECHNOLOGY
xv.	• GREEN BUILDING TECHNOLOGY: STABILIZED MUD BLOCK, TREATED BAMBOO, TERAFIL WATER, RAINWATER HARVESTING & LOW-COST TOILET • AGARBATI BAMBOO STICK • PRODUCTS FROM TREATED BAMBOO STICKS • COST-EFFECTIVE HATCHERY-SOLAR BASED	GREEN TECHNOLOGY, WATER, LIVELIHOOD, ENERGY
xvi.	• UP-GRADATION OF CONVENTIONAL RIVER ROPE WAY CROSSINGS OVER MAIN RIVERS EXISTS IN RURAL / HILLY TERRAINS	LIVELIHOOD
xvii.	• THIRUMATHIKART: E-COMMERCE PLATFORM FOR SHGS FOR ENHANCING THEIR LIVELIHOODS AND GENERATING NEW EMPLOYMENT OPPORTUNITIES.	
xviii.	• ECO-FRIENDLY AND BIO-DEGRADABLE POT BY USING COW DUNG AND AGRICULTURE WASTE • VALUE ADDITION OF CHARCOAL-BRIQUETTE BY ADDING LAC. • GOAT SPECIAL - CONCENTRATE MIXTURE PRODUCT	
xix.	• HERBAL MOSQUITO REPELLENT INCENSE STICK MAKING • HAND MADE PAPER PRODUCTION & PAPER BAG MAKING • CHALK PENCIL AND LIQUID DEODORANT PRODUCTS	
xx.	• CUSTOMIZATION IN BIOMASS PALLETISATION UNIT • TECHNOLOGY PROTOCOL FOR PRODUCTION OF FUEL PELLETS FROM BIOMASS • OPTIMIZATION PROCESS OF PELLET PRODUCTION USING LANTANA, AMLA PROCESSING WASTE, SUGARCANE BAGASSE, FOREST LEAVES, PADDY STRAW ETC.	GREEN TECHNOLOGY, LIVELIHOOD
xxi.	• HAND OPERATED SEWING MACHINE FOR DIFFERENTLY ABLED • COST EFFECTIVE AND ECO-FRIENDLY PAVER BLOCK MAKING • IMPROVED SILVER ANKLET PLATING TECHNOLOGY	HANDICRAFT, TEXTILE
xxii.	• FLEXIBLE CAGE FOR CATTLE TRANSPORT VEHICLE WITH LOADING ADJUSTMENT	ANIMAL HEALTH, LIVELIHOOD, AGRICULTURE
xxiii.	COMPACT AND EASILY MANOEUVRABLE TRACTOR OF LOW HP RANGE TO MEET THE REQUIREMENT OF THE SMALL FARMERS	
xxiv.	• POTABLE WATER USING FUNCTIONALISED CELLULOSE AS CATALYST FROM LOCALLY AVAILABLE BIOMASS	WATER
xxv.	WATER FILTRATION SOLUTIONS, BUILDING MATERIAL TECHNOLOGY SOLUTIONS	

xxvi.	• COST-EFFECTIVE NEEM FRUIT HARVESTER, AND PROCESS STANDARDIZATION • NEEM FRUIT COLLECTION AND POST-HARVEST PROCESSING TO PRODUCE QUALITY PULP AND NEEM SEED FOR PRODUCT DEVELOPMENT.	POST HARVEST
xxvii.	• PORTABLE LABORATORY WORKBENCH FOR LEARNING AND PROTOTYPING	EDUCATION

2. **Structural Engineering, Geotechnical Engineering, Geomatics, Construction Technology, and Management**
3. **Water and Environment Engineering, Transportation Engineering, Industrial and System Engineering**
4. **Energy Engineering, Energy for Sustainable Development, Waste Management**
5. **Data Science, Cyber Security, Computing, Pattern Recognition and Image Processing, Signal Processing**
6. **Big Data, IoT, Artificial Intelligence, Data Engineering, Robotics** (in agriculture, health, education, digital literacy, assistive aid)
7. **Networking and Communication Engineering, Embedded Systems, VLSI Design, Robotics**
8. **Power System Engineering, Electric Vehicles, Smart Grid Technologies, and Micro-grids**
9. **Materials and Design Engineering, Thermal and Fluid Engineering, Manufacturing Processes and Analysis**
10. **Control and Automation, Biomedical Engineering, Sensors and Measurement Techniques**
11. **Rural Engineering and Technical Support, Smart Villages and Infrastructure Development**
12. **Biological/Medical Science, Physical Science, Chemical Science, Agricultural Science, Biotechnology**
13. **Healthcare Management, Information Technology, Resources, and Environmental Management**

Key Areas of Focus

- **Artificial Intelligence (AI) and Robotics Technology:** Development and application of AI algorithms, machine learning models, and robotic systems for intelligent automation and decision-making.
- **5G Technology:** Exploring applications of 5G, including high-speed communication and low-latency services.

- **Internet of Things (IoT):** Creating interconnected devices and systems for data collection and exchange.
- **AI/ML and Augmented/Virtual Reality (AR/VR):** Combining AI with AR/VR for immersive experiences and enhanced real-world interaction.
- **Agriculture Innovations:** Smart farming techniques, precision agriculture, and agricultural drones for food security.
- **Smart Village Initiatives:** Projects to improve rural infrastructure and quality of life, including smart grids and telemedicine solutions.
- **Healthcare Technology:** Telemedicine platforms, AI for medical diagnosis, and wearable health monitors.
- **Digital Infrastructure:** Enhancing digital capabilities through secure platforms and improved connectivity.
- **Sustainable Engineering & Green Technologies:** Focus on renewable energy systems and eco-friendly material development.
- **Advanced Materials Science:** Research on novel materials with enhanced properties for various applications.
- **Cybersecurity & Data Analytics:** Developing security solutions and predictive models for data insights.
- **Mechatronics & Automation:** Integrating various engineering disciplines for advanced manufacturing and control systems.

For Master of Science (M.Sc.) Students

Depending on their specialization, M.Sc. students can explore areas such as:

- **Biotechnology & Bioinformatics:** Research in genetic engineering, drug discovery, agricultural biotechnology, or computational biology and genomics.
- **Environmental Science & Management:** Projects focusing on pollution control, ecological restoration, climate Modeling, sustainable resource management, or environmental impact assessment.
- **Physics & Applied Sciences:** Development of new sensors, advanced optical systems, quantum computing applications, or research in condensed matter physics.

For MBA Students

- **Digital Transformation & Business Analytics:** Developing strategies for digital business models, leveraging big data for market insights, customer relationship management (CRM) analytics, and optimizing supply chains through technology.

- **Sustainable Business Practices & ESG:** Creating business strategies focused on Environmental, Social, and Governance (ESG) factors, circular economy models, and corporate social responsibility.
- **FinTech & E-commerce Innovations:** Exploring new financial technologies, blockchain applications in business, digital payment systems, and innovative e-commerce strategies.

For MCA Students

- **Cloud Computing & DevOps:** Developing cloud-native applications, implementing CI/CD pipelines, optimizing cloud infrastructure, and exploring serverless architectures.
- **Web & Mobile Application Development:** Creating innovative and user-friendly web and mobile applications, focusing on UI/UX design, performance optimization, and cross-platform development.
- **Data Science & Machine Learning Applications:** Applying data science techniques to solve business problems, building predictive models, and developing machine learning solutions for various domains.

These areas offer diverse opportunities for students to engage in cutting-edge research and development.